# ORF309/MAT380
# Continuous Random Variables III: Independce

(pp. 74-76)

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# *Agenda for today*

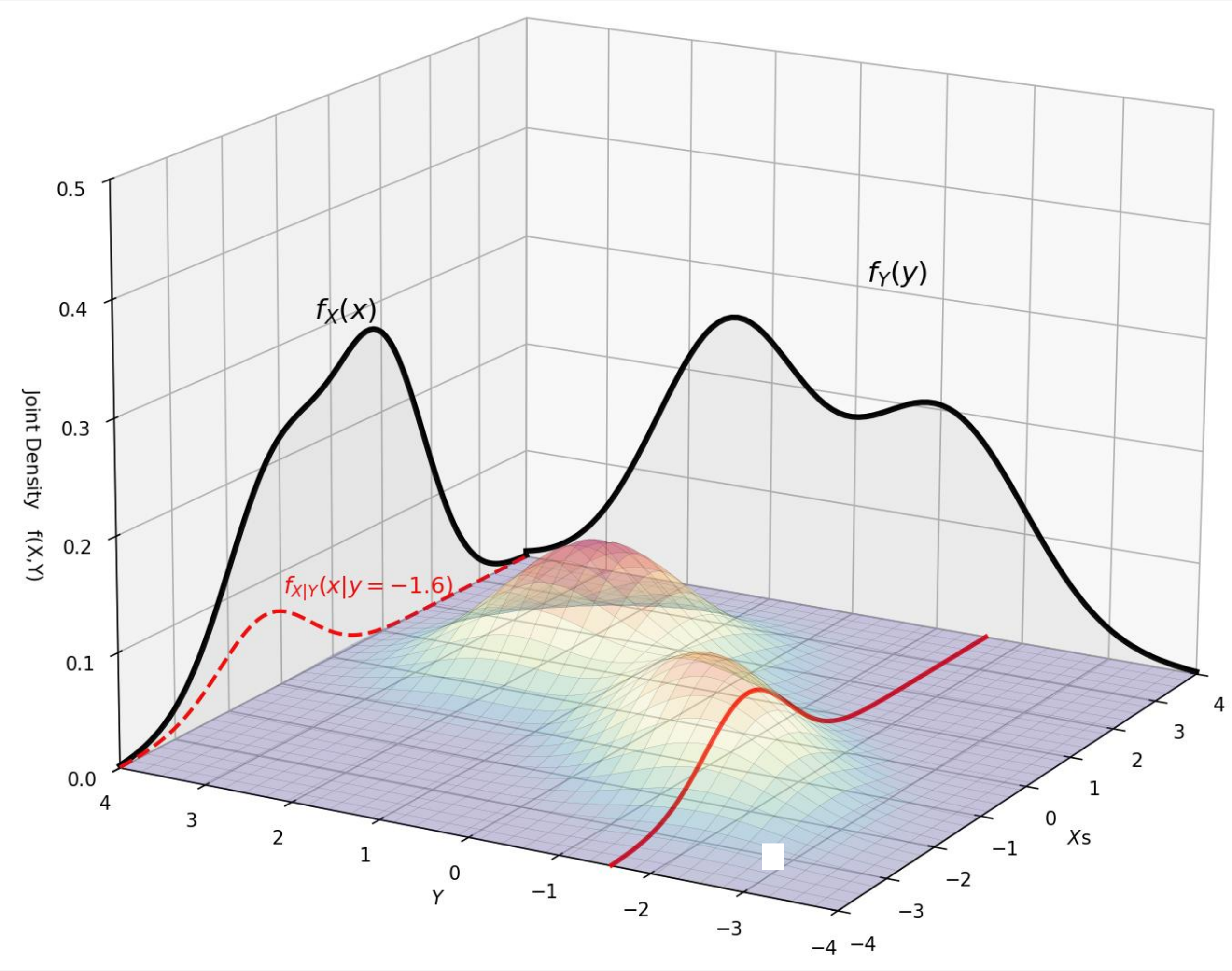
- Independence of CRVs
- Example

**Midterm 1 practice tests:**
**<u>now AVAILABLE on Canvas/Modules</u>**

**Midterm 1 will cover Chapters 1,2.**

**In-person exam with all the rules described in the announcement.**

This is a new plot showing the joint PDF, the corresponding marginal distributions and a conditional marginal distribution.



# *The importance of Independence*

- **Definition 1**: Two CRVs $X, Y$ are called independent ($X \perp\!\!\!\perp Y$) iff

$$P\{X \in dx \mid Y = y\} = P\{X \in dx\} \quad \Leftrightarrow \quad f_{X|Y}(x|y) = f_X(x), \forall x, y$$

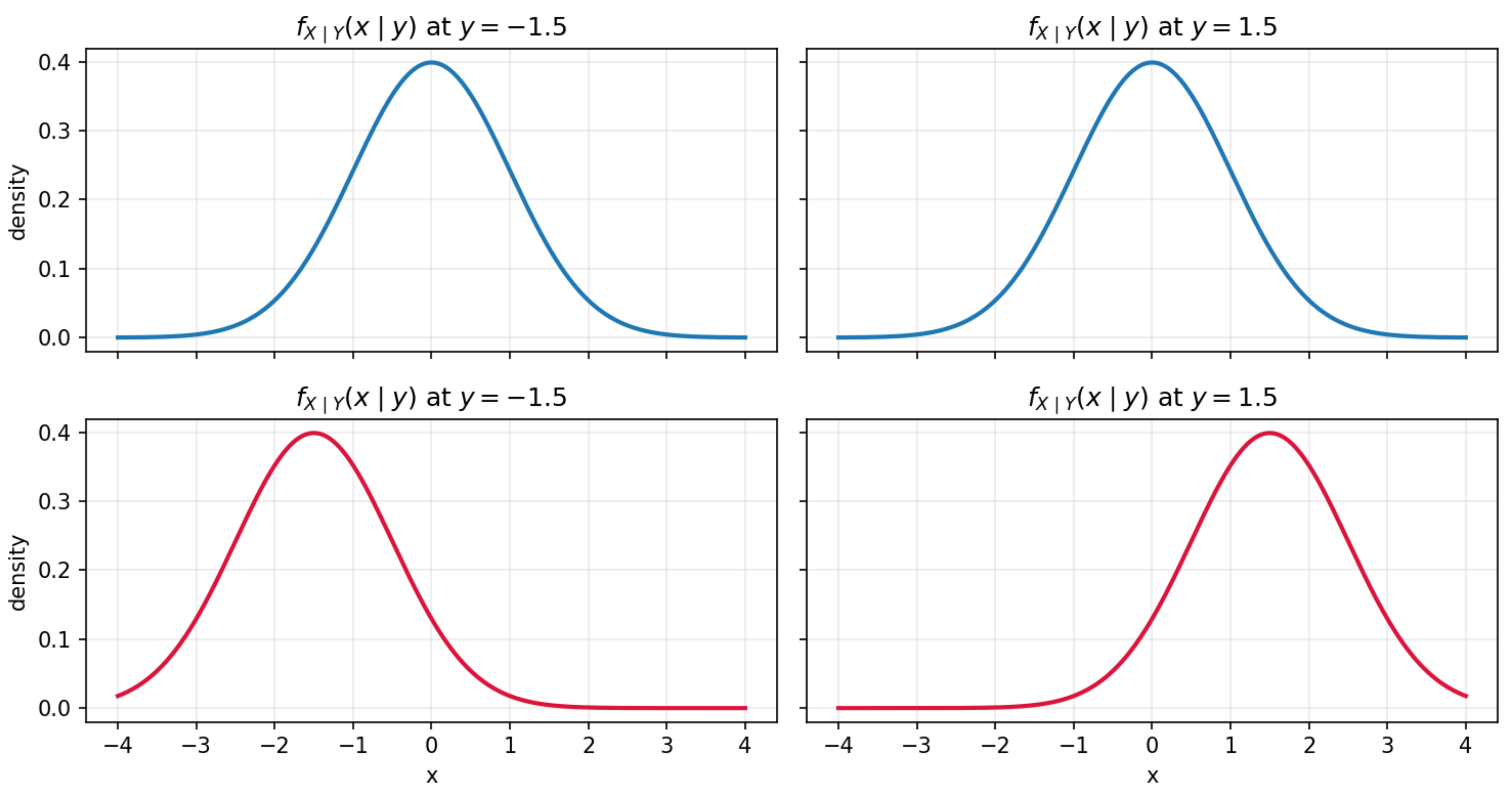
Equivalent Notations

- **Interpretation**: Conditioning on one of the variables does not impact the other

- **Real-world Intuition**: No matter what the **action** (*value*) of $\boldsymbol{Y}$ is, the **behavior** (*distribution*) of $\boldsymbol{X}$ is not affected → $\boldsymbol{X}$ **is a free spirit!**

$X, Y$ are independent

$X, Y$ are NOT independent



# *The importance of Independence*

- **“Algebraically massaging”** Definition 1, we can get another definition:

$$P\{X \in dx \mid Y = y\}\boldsymbol{P\{Y \in dy\}} = P\{X \in dx\}\boldsymbol{P\{Y \in dy\}} \Rightarrow$$

$$\Rightarrow P\{X \in dx, Y \in dy\} = P\{X \in dx\}\, P\{Y \in dy\}$$

- **Definition 2**: Two CRVs $X, Y$ are called independent $(X$  $Y)$ iff

$$P\{X \in dx, Y \in dy\} = P\{X \in dx\}\, P\{Y \in dy\}$$

- **Note**: if $X \perp\!\!\!\perp Y$ then $E[\varphi(X) \mid Y = y] = \int \varphi(x) P\{X \in dx \mid Y = y\} =$

$$= \int \varphi(x) P\{X \in dx\} = E[\varphi(X)]$$

# *Example*

- **Question**: How many people, (*on average*) will get on the 1st bus?
  - **Modeling assumptions**:
    - Passengers and buses arrive at a bus stop.
    - On average, $\mu$ passengers arrive at the bus stop in **1 hour**
    - On average, $\nu$ buses arrive at the bus stop in **1 hour**
    - The arrivals of passengers and buses are **independent**

## *Example*

- **Random variables ???**

# *Example*

- **Random variables:**
  - $T$ = the time the 1st bus arrives at the bus stop

    What is its distribution??

  - $N_t$ = the number of people waiting at the bus stop at time $t > 0$

    What is its distribution??

# *Example*

- **Random variables:**
  - $T$ = the time the 1st bus arrives at the bus stop
    - $T \sim Expon(\nu)$
  - $N_t$ = the number of people waiting at the bus stop at **time** $\boldsymbol{t} > 0$
    - $N_t \sim Pois(\mu t)$
  - $T \perp\!\!\!\perp N_t \ , \ \forall t$


Bus arrivals are **independent** of passenger arrivals.

# *Example*

- We know how to compute the expectation and distribution of $N_t$ for any time $t > 0$ .
- **But** we want to compute the expectation and distribution of $\boldsymbol{N_T}$
  - $T$ is a **random variable** and hence its value is **unknown**
- Let's think of **Tyche** and how she operates

## *Example*

- We know how to compute the expectation and distribution of $N_t$ for any time $t > 0$ .
- **But** we want to compute the expectation and distribution of $\boldsymbol{N_T}$
  - $T$ is a **random variable** and hence its value is **unknown**
- Let's think of **Tyche** and how she operates
  - In a **random experiment (i.e., some day)** **Tyche** will choose a specific outcome, say $\omega \in \Omega$
  - Once that $\omega$ is selected *randomness disappears*, and we have:
    - a specific time the bus arrives $T(\omega)$
    - a specific number of passengers $N_t(\omega)$ waiting at the bus stop at time $t$

# *Example*

- **Only Tyche can decide the <u>values</u> of** $N_t(\omega)$ **and** $T(\omega)$

  *Every day these values are different*

- We are interested in computing the **expectation** of $N_{T(\omega)}(\omega)$

  - This is the number of passengers, **on average**, waiting at the bus stop when the bus arrives

$$E[N_T] \;=\; \mathbf{??}$$

# *Example*

- **Only Tyche can decide $\omega$ and hence the <u>values</u> of** $N_t(\omega)$ **and** $T(\omega)$

  *Every day these values are different*

- We are interested in computing the **expectation** of $N_{T(\omega)}(\omega)$

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**Tower Property**

$$E[N_T] = \int_0^\infty E[N_T \mid T = t]\, P\{T \in dt\} =$$

# Example

- **Only Tyche can decide $\omega$ and hence the <u>values</u> of** $N_t(\omega)$ **and** $T(\omega)$

- We are interested in computing the **expectation** of $N_{T(\omega)}(\omega)$
  - This is the number of passengers, **on average**, waiting at the bus stop when the bus arrives

$$E[N_T] \overset{\text{Tower Property}}{=} \int_0^\infty E[N_T \mid T = t]\, P\{T \in dt\} \overset{T \perp\!\!\!\perp N_t}{=} \int_0^\infty \underset{N_t \sim Pois(\mu t)}{E[N_t]}\ \overset{T \sim Expon(\nu)}{P\{T \in dt\}} =$$

# *Example*

**Solution 1:**

$$E[N_T] \overset{\text{Tower Property}}{=} \int_0^\infty E[N_t \mid T = t]\, P\{T \in dt\} \overset{T \perp\!\!\!\perp N_t}{=} \int_0^\infty \underbrace{E[N_t]}_{N_t \sim Pois(\mu t)} \; \overset{T \sim Epon(\nu)}{P\{T \in dt\}} =$$

$$\int_0^\infty \mu t \;\; \nu e^{-\nu t} dt =$$

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$$\int_0^\infty \mu t\ \nu e^{-\nu t} dt = \mu\nu \int_0^\infty t e^{-\nu t}\, dt =$$

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$$\int_0^\infty \mu t\ \nu e^{-\nu t} dt = \mu\nu \int_0^\infty t e^{-\nu t}\, dt = -\mu\nu \frac{d}{d\nu}\left(\int_0^\infty e^{-\nu t}\, dt\right) =$$

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$$\boxed{E[N_T] = \frac{\mu}{\nu}}$$

*Example*

**Solution 2:**

$$E[N_T] \overset{\text{Tower Property}}{=} \int_0^\infty E[N_t \mid T = t]\, P\{T \in dt\} \overset{T \perp\!\!\!\perp N_t}{=} \int_0^\infty \underbrace{E[N_t]}_{N_t \sim Pois(\mu t)}\, \overbrace{P\{T \in dt\}}^{T \sim Epon(\nu)} =$$

$$\int_0^\infty \mu t\, P\{T \in dt\} =$$

# *Example*

**Solution 2:**

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$$\int_0^\infty \mu t\, P\{T \in dt\} = \mu \underbrace{\int_0^\infty t P\{T \in dt\}}_{\text{What does it remind you?}} =$$

## Example

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$$\int_0^\infty \mu t\, P\{T \in dt\} = \mu \underbrace{\int_0^\infty t P\{T \in dt\}}_{E[T]} =$$

# *Example*

**Solution 2:**

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$$\int_0^\infty \mu t\, P\{T \in dt\} = \mu \int_0^\infty t P\{T \in dt\} = \mu E[T] = \mu \frac{1}{\nu}$$

$$\boxed{E[N_T] = \frac{\mu}{\nu}}$$

*Calculus not required*

# *Example*

- Let's compute now the **distribution (PDF)** of the random variable $\boldsymbol{N_T}$

$$P\{\boldsymbol{N_T} = k\} = \int P\{N_T = k|T = t\}\, P\{T \in dt\} = \int P\{N_t = k|T = t\}\, P\{T \in dt\} =$$

# Example

- Let's compute now the **distribution (PDF)** of the random variable $\boldsymbol{N_T}$

$$P\{\boldsymbol{N_T} = k\} = \int P\{N_T = k | T = t\}\, P\{T \in dt\} = \int P\{N_t = k | T = t\}\, P\{T \in dt\} =$$

$$\overset{\perp\!\!\!\perp}{=} \int P\{N_t = k\}\, P\{T \in dt\} = \int \frac{e^{-\mu t}(\mu t)^k}{k!}\, \nu e^{-\nu t}\, dt = \int \frac{(\mu t)^k}{k!}\, \nu e^{-(\mu+\nu)t}\, dt = (\ldots) =$$

**(*k-times* integration by parts,**
which is impractical...**)**

# *Example*

- Let's compute now the **distribution (PDF)** of the random variable $\boldsymbol{N_T}$

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$$= \frac{\nu}{\mu+\nu}\left(\frac{\mu}{\mu+\nu}\right)^k \underbrace{\int \frac{(\mu+\nu)^{k+1} t^k\, e^{-(\mu+\nu)t}}{k!} dt}_{\text{Density of } Gamma(k+1,\mu+\nu)} = \frac{\nu}{\mu+\nu}\left(\frac{\mu}{\mu+\nu}\right)^k$$

**Proof in next pages…** →

## Let's now focus on the computations in (…)

**We will take a detour to see if we can avoid these integration by integrations**:

- Recall that $\boldsymbol{T_k \sim Gamma(k, \beta)}$ is a CRV that describes the **time of the $\boldsymbol{k}$-th arrival**

$$P\{T_k \leq t\} = 1 - \sum_{j=0}^{k-1} \frac{e^{-\beta t}(\beta t)^j}{j!}$$

- Let's find the density (**PDF**) of $\boldsymbol{Gamma(k, \beta)}$

$$P\{T_k \in dt\} = \frac{d}{dt}(P\{T_k \leq t\}) = \frac{d}{dt}\left(1 - \sum_{j=0}^{k-1} \frac{e^{-\beta t}(\beta t)^j}{j!}\right) = -\sum_{j=0}^{k-1} \frac{d}{dt}\left(\frac{e^{-\beta t}(\beta t)^j}{j!}\right) =$$

$$= -\sum_{j=0}^{k-1} \frac{d}{dt}\left(\frac{e^{-\beta t}(\beta t)^j}{j!}\right) = -\sum_{j=0}^{k-1}\left(\frac{-\beta e^{-\beta t}(\beta t)^j}{j!} + \frac{e^{-\beta t} j \beta (\beta t)^{j-1}}{j!}\right) =$$

$$= \sum_{j=0}^{k-1} \frac{\beta e^{-\beta t}(\beta t)^j}{j!} - \sum_{j=1}^{k-1} \frac{\beta e^{-\beta t}(\beta t)^{j-1}}{(j-1)!} = \cdots = \frac{\beta e^{-\beta t}(\beta t)^{k-1}}{(k-1)!}$$

same expression, but the counters differ by 1

Expanding the two summations and eliminating same terms yields…

**(only the <u>last term</u> of the <u>first summation</u> survives)**

Hence: $$P\{T_k \in dt\} = \frac{\beta e^{-\beta t}(\beta t)^{k-1}}{(k-1)!}$$

Recall also that we always have:

$$\int_0^\infty P\{T_k \in dt\} = 1 \Rightarrow \int_0^\infty \frac{\beta e^{-\beta t}(\beta t)^{k-1}}{(k-1)!} = 1$$

[End of Detour](#)

- Now let's go back to the **distribution** of $\boldsymbol{N_T}$ we had computed:

$$P\{\boldsymbol{N_T} = k\} = \frac{\mu^k \nu}{k!} \int t^k \, e^{-(\mu+\nu)t} dt \qquad (1)$$

- We will try to **convert** the red integral into a known distribution (note that we have $t^k$ and $e^{-(\mu+\nu)t}$ and $k!$)

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- We will try to **convert** the red integral into a <u>known distribution</u> (note that we have $t^k$ and $e^{-(\mu+\nu)t}$ and $k!$)

- This should remind us the PDF of the **Gamma distribution**

  - The $\boldsymbol{Gamma(k+1, \mu+\nu)}$ has density (substitute $\boldsymbol{\beta} = (\boldsymbol{\mu} + \boldsymbol{\nu})$ and $\boldsymbol{k} = \boldsymbol{k} + \mathbf{1}$ in the previous slide)

$$P\{T_{k+1} \in dt\} = \frac{d}{dt}(P\{T_{k+1} \leq t\}) = \frac{(\mu+\nu)e^{-(\mu+\nu)t}\big((\mu+\nu)t\big)^k}{k!} \qquad (2)$$

- Having (2) in our mind we can manipulate (1) so that to **convert** the red integral into an integral of the **density** of the $\boldsymbol{Gamma(k+1,\mu+\nu)}$ distribution:

$$\boldsymbol{P\{N_T = k\}} = \frac{\mu^k \nu}{k!} \int_0^\infty t^k e^{-(\mu+\nu)t} dt =$$

$$= \frac{\mu^k \nu}{k!} \frac{1}{(\mu+\nu)(\mu+\nu)^k} \int_0^\infty (\mu+\nu)(\mu+\nu)^k t^k e^{-(\mu+\nu)t} dt =$$

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$$= \frac{\mu^k \nu}{(\mu+\nu)(\mu+\nu)^k} \underbrace{\int_0^\infty \frac{\boldsymbol{(\mu+\nu)e^{-(\mu+\nu)t}\big((\mu+\nu)t\big)^k}}{\boldsymbol{k!}} dt}_{=1} =$$

Because of: $\boldsymbol{Gamma(k+1, \mu+\nu)}$

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**What does this remind you ??**

$$= \frac{\mu^k \nu}{(\mu+\nu)(\mu+\nu)^k} \underbrace{\int_0^\infty \frac{(\boldsymbol{\mu}+\boldsymbol{\nu}) e^{-(\mu+\nu)t} \left((\boldsymbol{\mu}+\boldsymbol{\nu})\boldsymbol{t}\right)^{\boldsymbol{k}}}{\boldsymbol{k}!} dt}_{=1} = \left(\frac{\boldsymbol{\mu}}{\boldsymbol{\mu}+\boldsymbol{\nu}}\right)^{\boldsymbol{k}} \frac{\boldsymbol{\nu}}{\boldsymbol{\mu}+\boldsymbol{\nu}}$$

Because of: $\boldsymbol{Gamma(k+1, \mu+\nu)}$

- Having (2) in our mind we can manipulate (1) so that to **convert** the red integral into an integral of the **density** of the $\boldsymbol{Gamma(k+1, \mu+\nu)}$ distribution:

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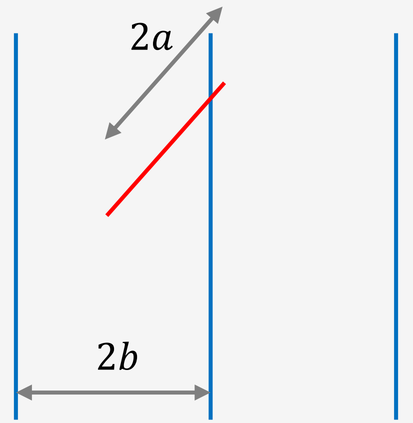
Because of: $\boldsymbol{Gamma(k+1, \mu+\nu)}$

$\boldsymbol{N_T + 1 \sim Geom(\nu/(\mu+\nu))}$

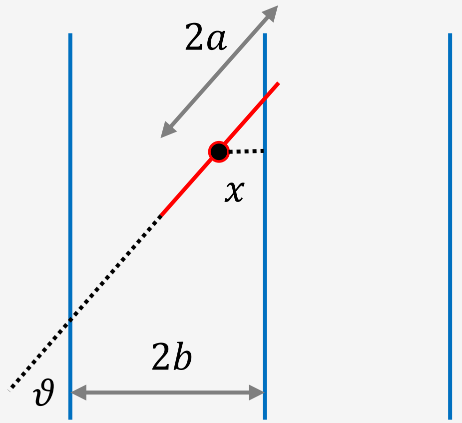
# Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

- **Example** (Buffon's needle – 1777):
- Perfect example for connecting **Geometry** and **Randomness**!
- Named after Georges-Louis Leclerc, Compte de Buffon (1707-1788) who was a French mathematician, cosmologist and naturalist.
- **The problem:**
  - You have a plane covered by parallel lines spaced a distance $2b$ apart.
  - You randomly drop a needle of length $2a$ onto the plane. Note: $a < b$.
  - **Buffon's question**: what is the probability that the needle **crosses/intersects** one of the parallel lines?



- The <u>event of interest</u> is: $\boldsymbol{A}$=**“the needle crosses on of the parallel lines”**.
- **Geometric intuition**:
  - A <u>*random drop*</u> of the needle is determined by
    - The <u>*distance*</u> $x$ of its middle point to the nearest line
    - The <u>*angle*</u> $\vartheta$ of the needle and the lines
  - Define the CRVs: $X \sim Unif([0,b])$ and $\Theta \sim Unif([0,\pi/2])$.
  - Also, $X$ and $\Theta$ are <u>**independent**</u> random variables
  - Then the **joint PDF** is: $f(x,y) = f_X(x) f_\Theta(\theta) = \frac{1}{b}\frac{2}{\pi}$ for all $(x,\theta) \in (0,b) \times (0,\pi/2)$.



- From **geometry** and using some **basic trigonometry** we can derive that the needle will **intersect** a parallel line for those $(x, \theta) \in (0, b) \times (0, \pi/2)$ such that

$$x \leq \mathrm{a} \cdot \cos(\theta)$$

  - **Recall that**: $\mathrm{a} \cdot \cos(\theta)$ is the orthogonal projection of the needle's half-length onto the *horizontal axis* (which is perpendicular to the parallel lines).

- Now the **event** $\boldsymbol{A}$=“the needle crosses on of the parallel lines” can be written equivalently by the event $\{\boldsymbol{X} \leq \mathbf{a} \cdot \boldsymbol{cos}(\boldsymbol{\Theta})\}$

  - Note that this is defined by two random variables…

- We need to compute the probability: $\boldsymbol{P}\{\boldsymbol{X} \leq \mathbf{a} \cdot \boldsymbol{cos}(\boldsymbol{\Theta})\}$ using **total probability.**

$$P\{X \le \mathrm{a} \cdot \cos(\Theta)\} = \int_0^{\pi/2} P\{X \le \mathrm{a} \cdot \cos(\theta) \mid \Theta = \vartheta\}\, P\{\Theta \in d\vartheta\} =$$

$$= \int_0^{\pi/2} P\{X \le \mathrm{a} \cdot \cos(\theta)\}\ f_\Theta(\theta) d\theta = \int_0^{\pi/2} \frac{1}{b} \mathrm{a} \cdot \cos(\theta) \frac{2}{\pi} d\vartheta =$$

$$= \frac{1}{b} \mathrm{a} \frac{2}{\pi} \int_0^{\pi/2} \cos(\theta)\, d\vartheta = \frac{2a}{\pi b}$$

- **<u>Remark</u>**: this probability can be used to **approximate the value of $\pi$.** Can you see how you could do that? Try to find yourselves before we see the solution in the next slide…

- Recall that the probability can be though of as the **fraction** of times an event happened when the number of trials is very large ($\to \infty$).

- Now suppose you throw the needle $\boldsymbol{n}$ times and $\boldsymbol{m}$ times ($\boldsymbol{m} \leq \boldsymbol{n}$) it intersects the parallel lines on the plane you have created

  - That event is: $A$ = "the needle crosses on of the parallel lines" = $\{X \leq \mathrm{a} \cdot \cos(\Theta)\}$.

- If $n \gg 0$ then you can argue: $P\{A\} \approx \frac{m}{n} \Rightarrow \frac{2a}{\pi b} \approx \frac{m}{n} \Rightarrow \pi \approx \frac{2an}{mb}$

  - Hence, as $n$ and $m$ become very large, we can approximate $\pi$ more accurately.

Who would ever imagine that Probability theory can be used to approximate $\pi$…